Advanced Database

Dr. M. Ali Nematbakhsh

Project3: Data Analytics

Section 1: Descriptive Questions

In this section, you must write complete descriptive answers based on your understanding. Answer one of the following two questions. Answering both will earn additional points.

Question 1: We have 3 Datasources with the following schemas:

- **Source A (CSV):** customer_id, name, phone, city
- **Source B (JSON):** user_id, order_date, total, city
- **Source C (SQL):** inv_no, cust_name, amt, date

Your Task:

1. **Design the Global Schema:** Define a single Target table schema (Attributes and Data Types) that captures all necessary data from these 3 Sources
2. **Map the Data:** Create a mapping Table Showing source attributes map to your target attributes
3. **Handle the Conflicts:** Source A uses customer_id and Source B uses user_id. Source C has no ID, only cust_name. Could you write a brief logical step describing how you will link a record from Source C to a corresponding record in Source A to prevent duplicate customer creation?

Question 2: You are a data warehouse designer for a large retail company.

You have 2 Tables:

- **Fact_Sales:** This is your fact table, which stores daily sales transactions. Example columns:
  Sale_ID, Date, Product_Key, Customer_Key, Quantity, Total_Amount
- **Product Information:** This will become your Product Dimension. You discover that products have a clear hierarchy: Department → Category → Brand → Product

Example of the raw, denormalized product data:

Product_Key	Product_Name	Brand_Name	Category_Name	Department_Name
1001	501 Jeans	Levi's	Pants	Men's Apparel
1002	T-Shirt	Nike	Tops	Men's Apparel
1003	Air Max	Nike	Shoes	Footwear
1004	501 Jeans	Levi's	Pants	Women's Apparel

Please draw a Star Schema diagram where Fact_Sales is the central table. It connects to one denormalized dimension table called Dim_Product.

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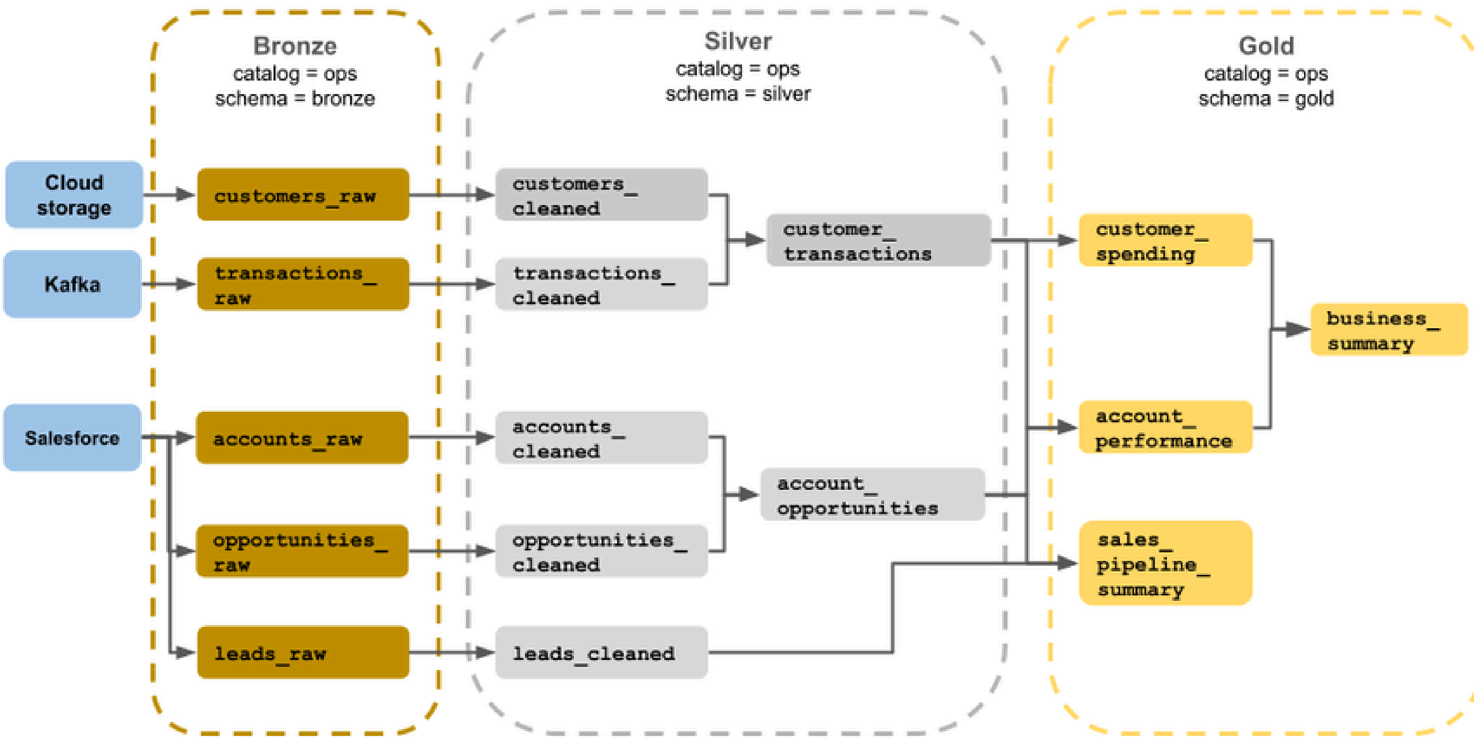
Section 2: Practical Questions
In this section, you'll face hands-on questions. This question is mandatory, unlike the previous section.

Why Medallion Architecture?
Modern data platforms like Databricks, Snowflake, and Google BigQuery rely on a layered data design known as the Medallion Architecture.
It's a best-practice framework for organizing data pipelines in three logical layers .



This approach helps organizations:

- **Ensure data quality and traceability from raw to refined.**
- **Make pipelines modular and maintainable.**
- **Enable fast analytics without compromising reliability.**

By mastering the Medallion pattern, you learn how modern data engineers design scalable, production-grade pipelines which is a skill used daily in real data engineering teams.

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You are a Data Engineer for Maven Roasters. Management needs a reliable data pipeline to analyze sales performance.

You have been tasked with implementing a Medallion Architecture using Python (Any Framework or Library like Pandas is Okay) to process raw sales data.

Input Data:
Use the provided Dataset: https://www.kaggle.com/datasets/ahmedabbas757/coffee-sales/data

Your Task:
Create a Jupyter Notebook that performs the following four steps. You must show your code and the output for the final analysis.

Step 1: The Bronze Layer (Raw Ingestion)
- Read the coffee_sales.csv file into a Pandas DataFrame.
- **Action:** Save this dataframe immediately as bronze_transactions.csv. This represents your immutable raw data.

Step 2: The Silver Layer (Cleaning & Enrichment)
- Load your Bronze data and apply the following transformations:
  a. Standardize Date: Create a new column transaction_timestamp by combining the date and time columns. Convert it to a proper datetime object.
  b. Calculate Metrics: Create a new column total_sale: Transaction_qty * unit_price
  c. Enrich: Extract the hour of the day from your timestamp into a new column named hour.
- **Action:** Save this cleaned DataFrame as silver_transactions.csv

Step 3: The Gold Layer (Aggregations)
- Using your Silver data, create 2 aggregated DataFrames optimized for reporting:
  a. df_product_sales: Group by product_type. Calculate the sum of total_sale (Revenue) and transaction_qty.
  b. df_store_performance: Group by store_location and transaction_date. Calculate the sum of total_sale (Daily Revenue).
- **Action:** Save these two tables as gold_product_sales.csv and gold_store_sales.csv.

Step 4: Business Analysis
Write a short query (code) on your Gold tables to print the answers to these three questions:
1. Which product_type generated the highest total revenue?
2. Which store location has the highest total revenue overall?
3. What is the average daily revenue for the "Astoria" location?

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What to upload for us?

- Answer all applicable questions (1 and/or 2).
- Your Notebook for Medallion Architecture
- All the CSV Files that you created for Medallion Architecture

All your files must be in one ZIP file named:
DBHW{Project Number}-{Student Complete name}-{Student Number}

Students must follow the Template structure, or they will receive a grade deduction.

Dead Line : 5 December - 14 Azar

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